

NIRMA UNIVERSITY
School of Technology, Institute of Technology
Master of Computer Application (2-Years Programme)
Semester-I

6CS101CC25 – Object Oriented Programming - List of Practical

Sr. No	Title	CLO	Marks	Hours
1	(data types, if..else and Switch) (A) Write a program in Java that calculate the radius of the circle by creating class circle. (B) Write a program in Java that will calculates the area of rectangle (C) Write a program that will display the grade of the student according to the percentage of the student using if...else ladder (D) Write a java program using switch statement which will display the name of the city according to the number used as input.	CLO-1	10	03
2	(Use of while, do..while and for loop) (A) Write a program in Java that calculate the radius of the circle by creating class circle. (B) Write a program in Java that will calculates the area of rectangle (C) Write a program that will calculate the factorial of the given Number. (D) Write a program to calculate the sum of the given number using Write program to produce the following output. <pre> 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5 </pre>	CLO-1	10	03
3	(String Handling) Write a program in java that shows the use of different methods available in String class .	CLO-2	10	02
4	(Constructor) (A) Write a Java program to create a class called "Dog" with a name and breed attribute. Create two instances of the "Dog" class, set their attributes using the constructor and modify the attributes using the setter methods and print the updated values. (B) Write a Java program to create a class called "Bank" with a collection of accounts and methods to add and remove accounts, and to deposit and withdraw money. Also define a class called "Account" to maintain account details of a particular customer.	CLO-2	10	04
5	(Static variables and methods) (A) Write a java program which will demonstrate the use of the static variable which is shared with all the objects. (B) Write a java program to get the cube of the given number using static method	CLO-2	10	02

6	Write a java program that shows the use of compile time and run-time polymorphism. (A) Write a Java program which will the concept of method overloading by creating two methods, first add() method performs addition of two numbers and second add() method performs addition of three numbers. (B) Write a Java program to create a class Vehicle with a method called speedUp(). Create two subclasses Car and Bicycle. Override the speedUp() method in each subclass to increase the vehicle's speed differently.	CLO-2	10	04
7	(A) Write a Java program to create an interface Shape with the getArea() method. Create three classes Rectangle, Circle, and Triangle that implement the Shape interface. Implement the getArea() method for each of the three classes. (B) Write a java program which will create a package and also import the package in the program.	CLO-3	10	04
8	(A) Write a Java program that throws an exception and catch it using a try-catch block. (B) Write a java program which will show the use of the custom exception in the program.	CLO-3	10	04
9	Write a Java program to create and start multiple threads that increment a shared counter variable concurrently.	CLO-3		02
10	Write a program in Java to copy one binary file to other using File Input Stream and Output Stream classes	CLO-3		02
11	Write Java code which will perform the insert operation on the Singer table according to the Singer id (Singer_id) and also write code to delete and update record from the Singer table. (Singer table fields: Singer_id, Singer_name and City).	CLO-3	Extra	
12	Assume that there are two different companies: one develops a student package and the other develops the staff package containing relevant classes. Implement a simple university system that makes use of classes/ capabilities provided by these two packages.	CLO-3	Extra	
Total				30

* Extra Practical